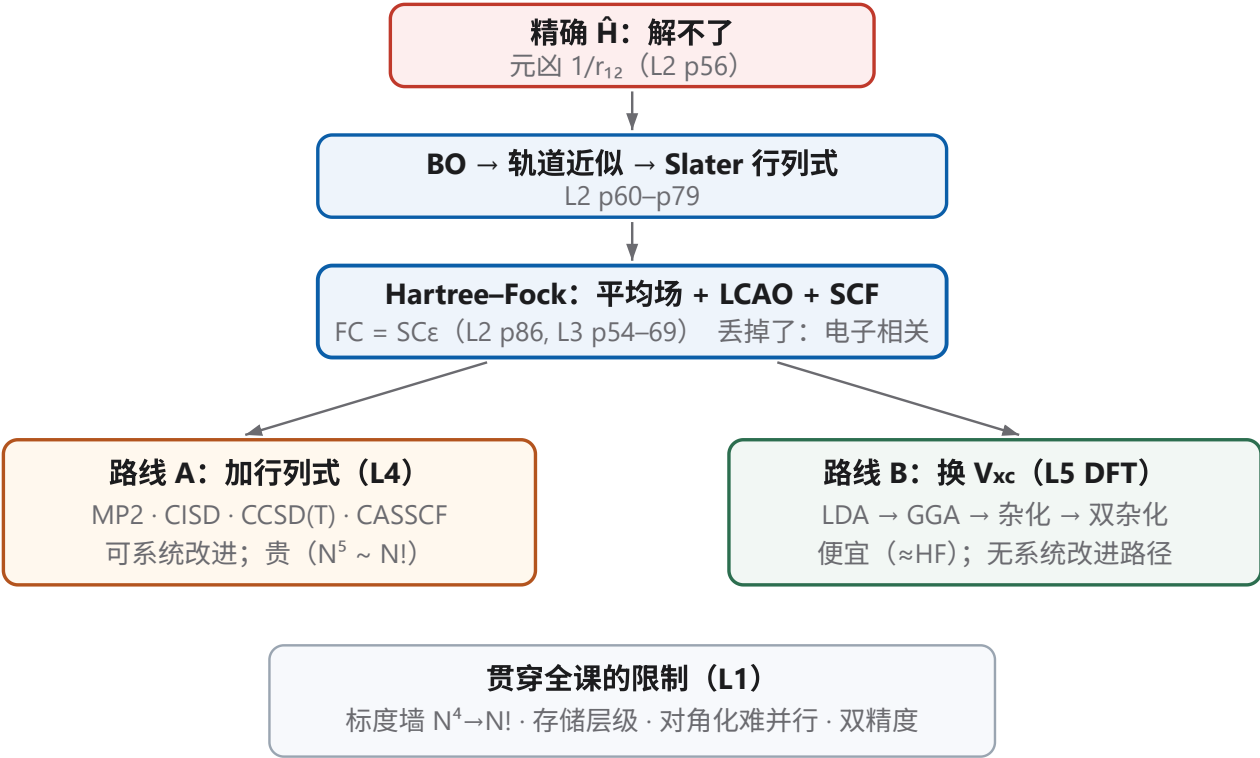


总汇总 第 1-6 讲

精简版：每讲 = 必背公式表 + 关键讲义原图（每图一句话）。看不懂的地方点"详解"跳到完整讲义对应页。

一张图看全课



第 1 讲 课程概览与计算平台

详解: [L01 完整讲义](#)

必背	一句话
考试形式	期中 15%、期末 25%， 只考理论/概念，不考计算 ；四次作业各 15%
四个软件	Gaussian 常规有机 · ORCA 光谱/磁性 · OpenMOLCAS 多组态 · VASP 固体
延迟 vs 带宽	延迟 = 第一字节多久到；带宽 = 之后多快到。RAM 100 周期，跨节点 10 ⁴ 周期
为什么双精度	总能量 ~10 ³ Hartree，要分辨 10 ⁻³ ，单精度 7 位全被吃光
矩阵分解	O(n ³)，内存不省、 难并行 → SCF 每轮对角化是瓶颈
Amdahl	S _{max} = 1/(1-p)，95% 可并行最多快 20 倍

摩尔定律的残酷	算力翻倍，CCSD(T) 能算的体系只大 10%——进步靠算法
集群三原则	别在登录节点跑；用 scratch； <code>%NProcShared = ncpus</code> 、 <code>%Mem < mem</code>

Scaling of elementary operations (BLAS & LAPACK):

Operation	CPU Cost / FLOPs	Memory cost	Sparse cost	Parallel execution
Matrix-matrix multiplication	$O(n^3)$	$O(nnz)$	LOWER	EASY
Matrix-vector multiplication	$O(n^2)$	$O(nnz)$	LOWER	EASY
exp(matrix)-vector multiplication	$O(n^2)$	$O(nnz)$	LOWER	EASY
Matrix exponentiation	$O(n^3)$	$O(nnz)$	LOWER	EASY
Most matrix factorizations	$O(n^3)$	$O(n^3)$	SAME	HARD

Matrix factorisations (SVD, LU, diagonalisation, Householder, etc.) are very expensive, difficult to parallelize, and do not benefit from sparse storage.

L1 p26 只有"矩阵分解"一行三项全差：内存 $O(n^3)$ 、稀疏没用、难并行。

第 2 讲 量子力学基础与 Hartree-Fock 理论

详解：[L02 完整讲义](#)

必背	一句话
何时需要量子力学	$\Delta E \gg k_B T$ (室温 ≈ 2.5 kJ/mol) ； 电子跃迁是 $100\text{--}400\ k_B T$
$\hat{H}\psi = E\psi$	本征值方程；量子化来自边界条件
$[\hat{x}, \hat{p}] = i\hbar$	不对易 $\rightarrow$ 不能同时确定 $\rightarrow$ 不确定性原理
厄米	本征值必为实 $\rightarrow$ 可观测量的算符必须厄米
Born 解释	$ \Psi ^2$ 是概率密度， Ψ 本身不可观测
五大公设	① Ψ 描述状态 ②可观测量 $\leftrightarrow$ 算符 ③测量值只能是本征值 ④期望值 $= \sum b_n ^2 a_n$ ⑤含时方程

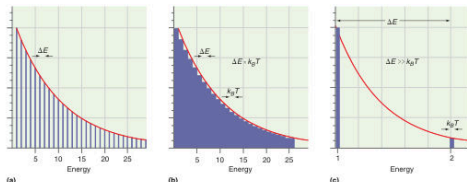
三个量子数	$[\hat{H}, \hat{L}^2] = [\hat{H}, \hat{L}_z] = 0$ (球对称) $\rightarrow n, l, m$; 节面总数 $n-1$
泡利	$\hat{P}_{12}\Psi = -\Psi$; Slater 行列式自动满足
J vs K	J : 经典库仑, 所有电子对都有; K : 纯量子, 仅同自旋, 降低能量
Hund 规则	$E(^3S) - E(^1S) = -2K$: 同自旋空间波函数反对称 $\rightarrow$ Fermi 空穴
BO 近似	核不动 $\rightarrow$ 势能面的概念由此而来
闭壳层 HF 能量	$E = 2\sum H_i + \sum \sum (2J_{ij} - K_{ij}) + V_{NN}$
Roothaan	FC = SCF ; S 因为基函数不正交; F 依赖 C $\rightarrow$ SCF 迭代
RHF vs UHF	H ₂ 解离: RHF 锁 50% 离子项 (高 ~ 140 kcal/mol); UHF 对但自旋污染

When should be quantum mechanics considered?

- Key concept: Boltzmann distribution:

$$\frac{n_i}{n_j} = \frac{g_i}{g_j} e^{-\frac{e_i - e_j}{k_B T}} = \frac{g_i}{g_j} e^{-\frac{\Delta E}{k_B T}}$$

Consider a quantum mechanical system with discrete energy levels



- If $\Delta E < k_B T$ or $\Delta E \approx k_B T$ $\Rightarrow$ spectrum appear to be continuous $\Rightarrow$ classical description is enough
- If $\Delta E > k_B T$ $\Rightarrow$ discrete spectrum $\Rightarrow$ quantum description is needed

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L2 p9 $\Delta E \lesssim k_B T$ 看起来连续 (经典够用); $\Delta E \gg k_B T$ 必须量子。

Hartree-Fock energy

- After some mathematical manipulation, the previous equation is simplified to (for closed shell molecules):

$$E = 2 \sum_i^{N_{elec}/2} H_i + \sum_i^{N_{elec}/2} \sum_j^{N_{elec}/2} (2J_{ij} - K_{ij}) + V_{NN}$$

Where:

$$\begin{aligned} H_i &= \langle \phi_i(1) | \hat{h}_i | \phi_i(1) \rangle && \text{= core Hamiltonian = monoelectronic integrals} \\ J_{ij} &= \langle \phi_i(1)\phi_j(2) | \hat{g}_{ij} | \phi_i(1)\phi_j(2) \rangle \\ K_{ij} &= \langle \phi_i(1)\phi_j(2) | \hat{g}_{ij} | \phi_j(1)\phi_i(2) \rangle && \text{= Electron Repulsion Integrals = bielectronic integrals} \end{aligned}$$

$$V_{NN} = \frac{1}{2} \sum_{\alpha} \sum_{\beta \neq \alpha}^{N_{nuc}} \frac{Z_{\alpha} Z_{\beta}}{|r_{\alpha} - r_{\beta}|}$$

83

L2 p83 闭壳层 HF 能量: 单电子积分 H + 双电子积分 $(2J-K)$ + 核排斥。

Hartree-Fock-Roothaan Equation

$$\sum_s^N c_{st} (F_{ts} - \epsilon_t S_{ts}) = 0, \quad t = 1, 2, 3, \dots, N$$

Where:

$$\begin{aligned} F_{ts} &= \langle \chi_t | \hat{F} | \chi_s \rangle \\ S_{ts} &= \langle \chi_t | \chi_s \rangle \end{aligned}$$

For HFR Equation to have a non trivial solution, the secular determinant must be zero:

$$|F_{ts} - \epsilon_t S_{ts}| = 0$$

Highlighted equations in the last several slides can be solved iteratively, using a self-consistent field (SCF) procedure to make the total energy convergent to a tolerant value. These equations form the basis of modern quantum & computational chemistry.

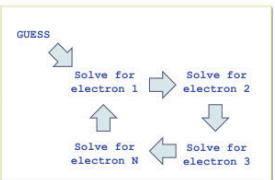
87

L2 p87 Roothaan: LCAO 把微分方程变成久期行列式 $|F - \epsilon S| = 0$ 。

Hartree-Fock-Roothaan Equation

Solving this for a single electron in an average electric field of all other electrons is relatively easy, and HF method proceeds by treating each electron in turn in a cycle (see the schematic on the right).

The algorithm updates the wavefunctions for each electron until the solution no longer changes. This procedure is known as the self-consistent field method.



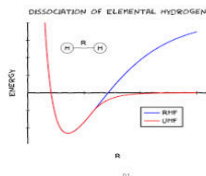
89

L2 p89 SCF: 猜 $\rightarrow$ 解 $\rightarrow$ 更新 $\rightarrow$ 重复到不再变化 (自治)。

Spin Restricted versus Spin Unrestricted

Spin-restricted: each occupied molecular orbital has two electrons. Spin-alpha density is exactly identical to spin-beta density. The shape of spin-alpha orbitals is identical to spin-beta orbitals.

Spin-unrestricted: Two sets of independent molecular orbitals are optimized. Spin-alpha and spin-beta orbitals. In large, at electronic ground state and equilibrium geometry, RHF and UHF are equivalent. However, UHF performs much better near dissociated limit. The shape of spin-beta orbitals is identical to spin-alpha orbitals. UHF wave function is subject to spin-contamination problem.



L2 p91 RHF α/β 共用轨道；UHF 两套独立轨道，解离好但有自旋污染。

第 3 讲 软件工具、变分原理、SCF 实现与基组

详解：[L03 完整讲义](#)

必背	一句话
变分原理	$E_{\text{trial}} \geq E_{\text{exact}}$ ；只保证能量，不保证性质和能量差
变分四步	写 $\psi(\alpha) \rightarrow$ 算 $E(\alpha) \rightarrow dE/d\alpha = 0 \rightarrow$ 代回
行列式能量（动笔题）	$E = \sum h_{ii} + \sum_{i < j} (J_{ij} - K_{ij})$ ；两个双占据轨道之间： $4J - 2K$
Koopmans	$IE \approx -\epsilon_{\text{HOMO}}$ ；冻结轨道；HF 误差抵消所以准；DFT 不适用
B 原子	2p 三重简并 $\rightarrow$ 需 3 个行列式 $\rightarrow$ 静态相关 $\rightarrow$ CASSCF
自旋污染	看 $\langle \hat{S}^2 \rangle$ vs $S(S+1)$ ：0 / 0.75 / 2.00
密度矩阵	$D_{\gamma\delta} = \sum_{\text{占据}} c_{\gamma i} c_{\delta i}$ ； $\text{tr}(\mathbf{D}\mathbf{S}) = N$
三种轨道	正则（ $\mathbf{F}$ 对角）·天然（ $\mathbf{D}$ 对角）·定域；能量密度全相同
双电子积分	$\sim N^4/8$ 个 $\rightarrow$ direct SCF 现算；RI 降到 N^3
收敛技巧	阻尼·能级移动（可能卡局部极小）·DIIS（误差 $\mathbf{FDS-SDF}$ ）·二次收敛
GTO vs STO	GTO 核处无尖点、衰减太快，但高斯乘积定理 $\rightarrow$ 积分解析
基组命名	6-311++G(3df,3pd)；aug-cc-pVnZ；极化 = $l+1$ ，弥散 = 阴离子/弱相互作用
BSSE	基函数互借 $\rightarrow$ 高估结合；Counterpoise + 幽灵原子
虚频	0 个 = 极小点；1 个 = 过渡态； ≥ 2 = 重做

Variational Principle

For any operator, when we use its own ground eigenstate – we get the lowest possible eigenvalue.

$$E_{\text{exact}} = \frac{\langle \psi_{\text{exact}} | \hat{H} | \psi_{\text{exact}} \rangle}{\langle \psi_{\text{exact}} | \psi_{\text{exact}} \rangle}$$

When we use an approximate function (i.e. not its eigenstate) to compute the expectation value of the operator – the result will always be higher.

$$E_{\text{trial}} = \frac{\langle \psi_{\text{trial}} | \hat{H} | \psi_{\text{trial}} \rangle}{\langle \psi_{\text{trial}} | \psi_{\text{trial}} \rangle}$$

The above theorem states that: $E_{\text{trial}} \geq E_{\text{exact}}$

Proof:

Expand the trial function as a linear combination of exact eigenstates of the Hamiltonian (= decompose ψ_{trial} in the basis set of the Hamiltonian):

$$\psi_{\text{trial}} = \sum_{n=1} c_n \varphi_n, \quad \text{where} \quad \hat{H} \varphi_n = \varepsilon_n \varphi_n$$

$$E_{\text{trial}} = \langle \psi_{\text{trial}} | \hat{H} | \psi_{\text{trial}} \rangle = \sum_{n=1} \sum_{m=1} c_n^* c_m \langle \varphi_n | \hat{H} | \varphi_m \rangle = \sum_{n=1} \sum_{m=1} c_n^* c_m \langle \varphi_n | \varepsilon_m \varphi_m \rangle =$$

$$= \sum_{n=1} \sum_{m=1} c_n^* c_m \varepsilon_n \langle \varphi_n | \varphi_n \rangle = \sum_{n=1} \sum_{m=1} c_n^* c_m \varepsilon_n \delta_{mn} = \sum_{n=1} c_n^* c_n \varepsilon_n$$

On the other hand:

$$E_{\text{exact}} = \langle \psi_{\text{exact}} | \hat{H} | \psi_{\text{exact}} \rangle = \langle \varphi_1 | \hat{H} | \varphi_1 \rangle = \varepsilon_1$$

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L3 p26 变分原理证明：展开后 $\sum |c_i|^2 (\varepsilon_i - \varepsilon_1) \geq 0$ 。

Energy of a Slater Determinant

General energy expression for a given Slater Determinant:

$$E = \sum_i^{N_{\text{elec}}} h_{ii} + \sum_i \sum_{j>i}^{N_{\text{elec}}} (J_{ij} - K_{ij}) + V_{\text{nn}}$$

J_{ij} – is non-zero for all pairs of electrons, regardless of their spin

K_{ij} – is non-zero for all pairs of electrons with same spin

Calculate the energy of the following Slater Determinants:

$$\begin{array}{c} \uparrow \downarrow \\ \uparrow \downarrow \\ \uparrow \end{array} \begin{array}{c} 3 \\ 2 \\ 1 \end{array} \quad E = 2h_{11} + h_{22} + h_{33} + J_{11} + 2J_{12} + 2J_{13} + J_{23} - K_{12} - K_{13} - K_{23}$$

$$\begin{array}{c} \uparrow \downarrow \\ \uparrow \downarrow \\ \uparrow \downarrow \end{array} \begin{array}{c} 3 \\ 2 \\ 1 \end{array} \quad E = 2h_{11} + 2h_{22} + J_{11} + J_{22} + 4J_{12} - 2K_{12}$$

44

L3 p44 数电子对：同自旋才有 $-K$ 。唯一的动笔题型。

Koopmans Theorem

Let's use the obtained set of molecular orbitals for a system with N electrons to compute the energy of a system with (N+1) and (N-1) electrons. Consider that the orbitals remain exactly the same as in the original calculation.

N-1 case

$$E_{N-1} = 2h_{11} + h_{22} + J_{11} + 2J_{12} - K_{12}$$

N case

$$E_N = 2h_{11} + h_{22} + h_{33} + J_{11} + 2J_{12} + 2J_{13} + J_{23} - K_{12} - K_{13} - K_{23}$$

N+1 case

$$E_{N+1} = 2h_{11} + h_{22} + h_{33} + h_{44} + J_{11} + 2J_{12} + 2J_{13} + 2J_{14} + J_{23} + J_{24} + J_{34} - K_{12} - K_{13} - K_{14} - K_{23} - K_{24} - K_{34}$$

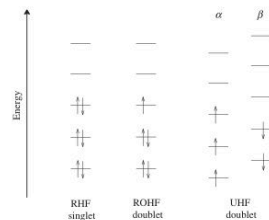
47

L3 p47 Koopmans: N 、 $N \pm 1$ 用同一套轨道，能量差 = 轨道能。

Restricted and Unrestricted SCF

• The Hamiltonian used in Hartree-Fock theory does not explicitly mention spin, and therefore, has a symmetry with respect to spin – a spin-flip does not change the energy. If we enforce that symmetry (demanding that the up and the down spin orbitals share the space part), the version of the Hartree-Fock (and more generally SCF) method that results is called *restricted*.

• In open-shell systems, the *unrestricted* version is preferred, where the two space parts are allowed to differ.

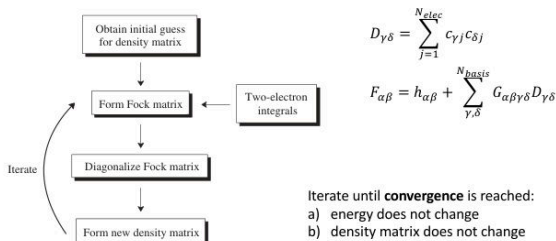


51

L3 p51 RHF / ROHF / UHF 的占据方式。

SCF Procedure

$$E = \sum_{\alpha, \beta}^{N_{\text{basis}}} D_{\alpha\beta} h_{\alpha\beta} + \frac{1}{2} \sum_{\alpha, \beta, \gamma, \delta}^{N_{\text{basis}}} D_{\alpha\beta} D_{\gamma\delta} (\langle \chi_{\alpha} \chi_{\gamma} | \mathbf{g} | \chi_{\beta} \chi_{\delta} \rangle - \langle \chi_{\alpha} \chi_{\gamma} | \mathbf{g} | \chi_{\delta} \chi_{\beta} \rangle) + V_{\text{nn}}$$

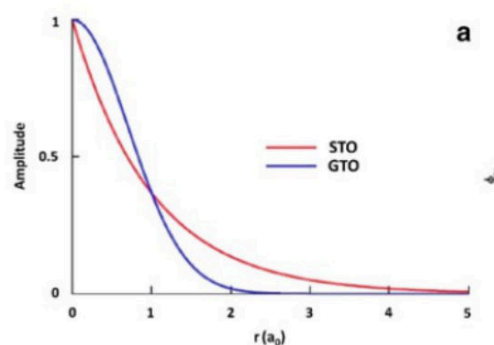


Iterate until **convergence** is reached:
a) energy does not change
b) density matrix does not change

Variational Principle!

60

L3 p60 SCF 矩阵版：猜 **D** → 建 **F** → 对角化 → 新 **D**。

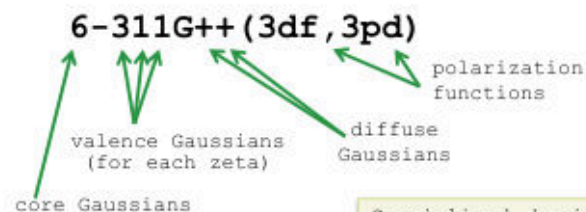


Comparison of Slater- and Gaussian-type orbitals is shown schematically. The orbital exponent for both the orbitals is 1.0. Slater type orbital can be “fitted” using several Gaussian functions, with different exponents and radii.

L3 p72 STO 核处尖、长程慢衰减；GTO 核处平、衰减太快 → 用几个 GTO 拼一个 STO。

Gaussian Basis Sets

Pople's basis sets:



Correlation-consistent basis sets:

aug-cc-pVTZ

"correlation consistent polarized valence triple-zeta basis set augmented with diffuse functions"

What to look for:

Specialized basis sets exist for almost any specific purpose. Check the literature for any case-specific basis sets before using a general purpose one.

Zeta number	Polarization functions	Diffuse functions	Tight functions	Effective core potentials	Rydberg functions
The more, the better/slower, usually 2-3.	The more, the better/slower, Usually one or two $l+1$ sets.	The more, the better/slower, usually a single set for anions.	Only used for computing HFCs and J -couplings.	Used to approximate inner shells of heavy elements	Only used to describe Rydberg states

Bigger and more flexible basis sets provide more accurate matrix representations for the original differential equations and therefore yield more accurate results.

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L3 p76 Pople 与 Dunning 基组命名拆解。

第 4 讲 电子相关方法

详解: [L04 完整讲义](#)

必背	一句话
相关能	$E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}} < 0$; $\sim 1\%$ 总能量, 但与键能同量级
静态 vs 动态	静态: 单行列式原理上不够 (简并/断键) $\rightarrow$ CASSCF; 动态: 瞬时躲避 $\rightarrow$ MP2/CC
Brillouin	$\langle \Psi_{\text{HF}} \hat{H} \Psi_i^r \rangle = 0 \rightarrow$ 单激发不直接混入、MP2 无单激发项
Ne 权重	参考 $0.96 \cdot$ 双激发 $0.034 \cdot$ 单激发 $0.001 \rightarrow$ 双激发是主力
CI 修 H_2	$\Psi = (a_0 + a_1)[\text{离子}] + (a_0 - a_1)[\text{共价}]$; $R \rightarrow \infty$ 时 $a_1 = -a_0$
尺寸一致性	截断 CI $\times$ (缺 $T_A T_B$ 交叉项); CC $\checkmark$ ($e^{T_A + T_B} = e^{T_A} e^{T_B}$)
CC	$\Psi = e^T \Psi_0$; $\frac{1}{2} T_2^2$ 隐含四激发; 非变分; CCSD(T) 黄金标准

MP2	$\Sigma \langle ij ab \rangle ^2 / (\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b)$; 简并时分母 $\rightarrow 0$ 发散
MPn 不收敛	Be: MP7 = 100.15%, 会过冲
标度	HF N^4 · MP2 N^5 · CCSD N^6 · CCSD(T) N^7 · FCI $N!$
CBS	HF 指数收敛; 相关能 A/n^3 慢收敛 (电子尖点)

Electron Correlation

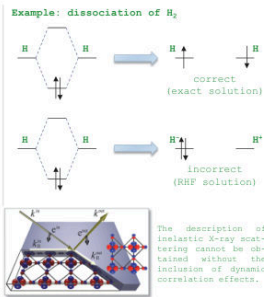
Electron correlation is, in effect, defined as inter-electron interactions that Hartree-Fock theory does not properly account for.

Static correlation refers to a situation where a single Slater determinant (*i.e.* a single population pattern of molecular orbitals) might not be a good approximation to the ground state. This often happens when the ground state is or becomes degenerate.

Example: dissociation of H_2 :

A famous example is the dissociation of the hydrogen molecule, where the restricted Hartree-Fock method produces a pair of ions instead of a pair of radicals.

Dynamic correlation is the correlated *motion* of electrons resulting from the Coulomb repulsion. Being an average field theory, the HF method does not properly account for this effect.



4

L4 p4 静态相关 (RHF 把 H_2 拆成 $H^- + H^+$) vs 动态相关。

Weight of excitation level for Ne in [5s4p3d] basis

Excitation level	Weight
0	9.6×10^{-1}
1	9.8×10^{-4}
2	3.4×10^{-2}
3	3.7×10^{-4}
4	4.5×10^{-4}
5	1.9×10^{-5}
6	1.7×10^{-6}
7	1.4×10^{-7}
8	1.1×10^{-9}

16

L4 p16 激发级别权重: 双激发远大于单激发, 4 级以上断崖下跌。

CI matrix for H_2 in minimal basis

As result, the diagonalization of the CI matrix leads to a mixing between Ψ_0 (ground state at HF) and doubly-excited Ψ_1 .

$$\begin{aligned} \Psi_0 &= \begin{vmatrix} \varphi_g(1) & \bar{\varphi}_g(1) \\ \varphi_g(2) & \bar{\varphi}_g(2) \end{vmatrix} = \varphi_g(1)\bar{\varphi}_g(2) - \bar{\varphi}_g(1)\varphi_g(2) = \varphi_g(1)\varphi_g(2)(\alpha\beta - \beta\alpha) = \\ &= (\chi_A(1) + \chi_B(1))(\chi_A(2) + \chi_B(2)) = \underbrace{\chi_A(1)\chi_A(2) + \chi_B(1)\chi_B(2)}_{\text{ionic terms}} + \underbrace{\chi_A(1)\chi_B(2) + \chi_B(1)\chi_A(2)}_{\text{covalent terms}} \end{aligned}$$

$$\begin{aligned} \Psi_1 &= \begin{vmatrix} \varphi_u(1) & \bar{\varphi}_u(1) \\ \varphi_u(2) & \bar{\varphi}_u(2) \end{vmatrix} = \varphi_u(1)\bar{\varphi}_u(2) - \bar{\varphi}_u(1)\varphi_u(2) = \varphi_u(1)\varphi_u(2)(\alpha\beta - \beta\alpha) \\ &= (\chi_A(1) - \chi_B(1))(\chi_A(2) - \chi_B(2)) = \chi_A(1)\chi_A(2) + \chi_B(1)\chi_B(2) - \chi_A(1)\chi_B(2) - \chi_B(1)\chi_A(2) \end{aligned}$$

The Full CI is a linear combination between Ψ_0 and Ψ_1 :

$$\Psi_{FCI} = a_0\Psi_0 + a_1\Psi_1, \text{ with } a_0^2 + a_1^2 = 1$$

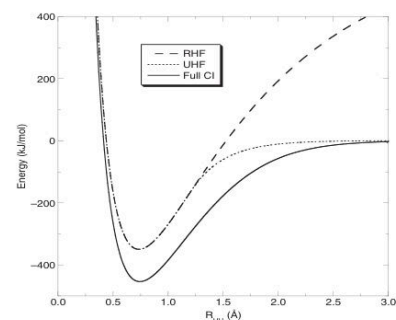
$$\Psi_{FCI} = (a_0 + a_1)(\chi_A(1)\chi_A(2) + \chi_B(1)\chi_B(2)) + (a_0 - a_1)(\chi_A(1)\chi_B(2) + \chi_B(1)\chi_A(2))$$

The HF wave function Ψ_0 has 50% ionic and 50% covalent contributions. By admixing the Ψ_1 into Ψ_0 , the **covalent** contribution usually increases.

19

L4 p19 HF 行列式 = 50% 离子 + 50% 共价; 混入 σ_u^2 后比例可调。

Comparison between RHF, UHF and FCI for H_2



21

L4 p21 全课最重要的一张图: RHF 长程灾难、UHF 能量对、FCI 精确。

Size Inconsistency of truncated CI methods

Let's assume we need to compute the energy of two water molecules, separated by e.g. 100

Angstrom. We can perform 2 calculations:

- 1) Compute *both water molecules together*, in the same calculation (assign basis sets to all 2O and 4 H) → get total energy E_{full} .
- 2) Compute *one single water molecule* and then, multiply the result by 2. Get the total energy E_{sum} .

We would expect that $E_{full} = E_{sum}$. However, this equality is respected only by the Full CI method (when all excitations are included in both calculations).

When using any of the truncated CI methods (like CISD), we will find that $E_{sum} < E_{full}$. The reason for this behavior arises from the fact that the E_{full} included only single and double excitations for the entire basis set of both molecules, while E_{sum} includes single and double excitations for each fragment only. When we multiply by 2 the energy of the one water molecule, we essentially take the product of two CISD wave functions of each water molecule. Such a product of two CISD wave functions will necessarily include single, double, triple and quadruple excited SDs, more than the CISD wave function obtained for the case (1).

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L4 p22 两个相距 100 Å 的水: CISD 下 $E_{sum} < E_{together}$

Coupled Clusters (CC)

Coupled Clusters method (like CI) aims at adding excited Slater Determinants into the ground state HF wave function. The difference between CC and CI lies in the selection of which excited SD are included in the wave function.

$$T = T_1 + T_2 + T_3 + \dots + T_{N_{elec}}$$

T_1 —operator allowing to generate singly-excited configurations

T_2 —operator allowing to generate doubly-excited configurations

The CI wave function can be written:

$$\Psi_{CI} = (1 + T)\Psi_0 = (1 + T_1 + T_2 + T_3 + \dots + T_{N_{elec}})\Psi_0$$

The corresponding Coupled Clusters wave function is defined as:

$$\Psi_{CC} = e^T \Psi_0 \Rightarrow (\text{Taylor expansion of } e^T)$$

$$= \left[1 + T_1 + \left(T_2 + \frac{1}{2} T_1^2 \right) + \left(T_3 + T_2 T_1 + \frac{1}{6} T_1^3 \right) + \dots \right] \Psi_0$$

By comparison with the CI wave function, the CC wave function at each excitation level contains additional terms arising from products of excitations. Schrodinger equation becomes:

$$H\Psi_{CC} = E_{CC}\Psi_{CC} \rightarrow H e^T \Psi_0 = E_{CC} e^T \Psi_0$$

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L4 p24 $CI = (1+T)\Psi_0$; $CC = e^T\Psi_0$, 展开多出乘积项。

Moller-Plesset Perturbation theory

It is the straightforward application of PT to the closed shell HF wave function:

$$\psi_n^{(0)} = \Psi_0 = \Psi_{HF}$$

$$E \approx E(MP0) + E(MP1) + E(MP2) = E_0 + \sum_{i<j} \sum_{a<b}^{occ} \frac{(\langle \varphi_i \varphi_j | \varphi_a \varphi_b \rangle - \langle \varphi_i \varphi_j | \varphi_b \varphi_a \rangle)^2}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}$$

$\varepsilon_i, \varepsilon_j$ —energies of the occupied orbitals in HF wave function

$\varepsilon_a, \varepsilon_b$ —energies of the empty orbitals in HF wave function

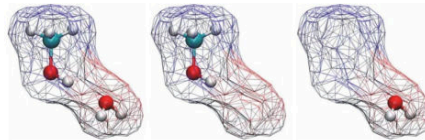
36

L4 p36 MP2 公式: 只含双激发, 分母是轨道能差。

Counterpoise Correction to BSSE

Atom-centred basis sets are rarely big enough to avoid the basis set superposition error. Counterpoise correction aims to compensate this error.

1. The energy of the complex is calculated in the usual way.
2. Two additional calculations are carried out for each fragment in the presence of orbitals (but not electrons or nuclei) of the other fragment.



$$\Delta E_{CP} = E(A)_{AB} + E(B)_{AB} - E(A)_A - E(B)_B$$

Basis Set	H_0 (no CP correction) (in hartree)	Counterpoise corrected energy (in hartree)	Counterpoise BSSE Energy (in kcal/mol)
3-21G	-151.1894	-151.13462	4.066
6-31G	-151.9342	-151.93274	0.916
6-31+G(d)	151.99457	-151.99364	0.584

L4 p38 Counterpoise: 三次计算都用完整 AB 基组。

Scaling of explicit correlation methods

Methods (without basis screening)	Scaling with the number of basis functions
HF	$O(n^4)$
MP2	$O(n^5)$
MP3, CISD, CCSD, QCISD	$O(n^6)$
MP4, CCSD(T), QCISD(T)	$O(n^7)$
MP5, CISDT, CCSDT	$O(n^8)$
MP6	$O(n^9)$
MP7, CISDTQ, CCSDTQ	$O(n^{10})$
FCI	$O(n!)$

Hartree-Fock:

- Up to a few hundred atoms.
- First and second analytical derivatives readily available with respect to common perturbations.

MP2, MP4:

- Efficient implementations scale to ~100 atoms for MP2 and to ~20 atoms for MP4.
- First and second analytical derivatives readily available with respect to common perturbations for MP2, numerical derivatives only for MP4.

CISD, CCSD:

- Efficient implementations scale to ~20 atoms.
- First analytical derivatives usually available, second derivatives mostly numerical for now.

FCI:

- 3-4 simple atoms at most.
- All derivatives numerical.

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L4 p40 各方法的标度与可算体系大小。

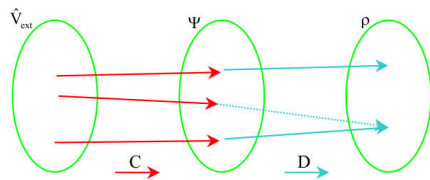
第 5 讲 密度泛函理论 (DFT)

详解: [L05 完整讲义 \(每页附原图\)](#)

必背	一句话
密度	$\rho(\mathbf{r})$ 只有 3 个变量, 与 N 无关; $N = \sum P_{\mu\nu} S_{\mu\nu}$
外势	$\hat{T}$, $\hat{V}_{ee}$ 普适, 只有 V_{ext} 区分分子
HK 第一定理	基态 $\rho \Leftrightarrow V_{\text{ext}}$, 一切性质是 ρ 的泛函 (反证法证明)
HK 第二定理	$E[\rho] = F_{\text{HK}}[\rho] + \int \rho V_{\text{ext}}$, 在 ρ_0 处最小
泛函	函数 $\rightarrow$ 数; $\delta(\int \rho v) / \delta \rho = v$
Kohn-Sham	无相互作用替身重现 ρ ; $[-\frac{1}{2} \nabla^2 + V_{\text{H}} + V_{\text{ext}} + V_{\text{xc}}] \varphi = \epsilon \varphi$; 只有 V_{xc} 未知
HF vs KS	HF: 交换精确非局域、无相关; KS: 交换+相关都近似但局域
Jacob 阶梯	$\text{LDA}(\rho) \rightarrow \text{GGA}(\nabla \rho) \rightarrow \text{meta-GGA}(\tau) \rightarrow \text{杂化(HF 交换)} \rightarrow \text{双杂化(MP2)}$

B3LYP	20% HF 交换, 3 个经验参数 → 严格说非 ab initio
SIE	HF 中 $J_{ii}=K_{ii}$ 抵消; 近似 DFT 抵消不净 → 过度离域、能垒低、带隙小
范围分离	erf 拆 $1/r_{12}$, 长程 HF 交换 → 修 CT 激发、SIE
色散	局域泛函看不到远处 → D3: $-\Sigma C_6/R^6 \times$ 阻尼
不适合 DFT	双自由基、开壳层过渡金属、断键; 镧系用 4f-in-core
磁耦合	$\hat{H} = -2J\hat{S}_1\cdot\hat{S}_2 \rightarrow E_T = -J/2, E_S = 3J/2, \Delta E_{ST} = -2J$
BS-DFT	单重态需 2 个行列式 → 算 $ \alpha\beta\rangle$; $J = (E_{BS}-E_{HS})/(\langle S^2 \rangle_{HS}-\langle S^2 \rangle_{BS})$
TD-DFT	线性响应, 极化率极点 = 激发能; 绝热近似; CT/Rydberg/双激发差

First Theorem of Hohenberg and Kohn



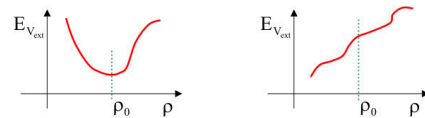
$C: \hat{V}_{\text{ext}} \mapsto \Psi$ injective (no surprise)
 $D: \Psi \mapsto \rho$ injective (very surprising!)
 = first theorem of Hohenberg and Kohn (1964)

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L5 p14 HK 第一定理: $\Psi \rightarrow \rho$ 是单射 ("非常意外")。

Second Theorem of Hohenberg and Kohn

"The unique functional that returns the ground state total energy when applied to the ground state density, returns a higher energy for any other density."



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L5 p19 HK 第二定理: 精确泛函在 ρ_0 处取最小 (左图)。

Theorem of Kohn and Sham

"The exact ground state density of a N -electron system is

$$\rho(\vec{r}) = \sum_{i=1}^N \phi_i^*(\vec{r}) \phi_i(\vec{r})$$

where the single-particle wave functions ϕ_i are the N lowest-energy solutions of the Kohn-Sham equations:

$$\begin{aligned} \hat{H}_{\text{KS}} \phi_i &= \epsilon_i \phi_i \\ \text{with } \hat{H}_{\text{KS}} &= \hat{T}_0 + \hat{V}_H + \hat{V}_{\text{ext}} + \hat{V}_{\text{xc}} \\ &= -\frac{\hbar^2}{2m} \nabla^2 + \frac{e^2}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} d\vec{r}' + \hat{V}_{\text{ext}} + \hat{V}_{\text{xc}} \\ \text{and } \hat{V}_{\text{xc}} &= \frac{\delta V_{\text{xc}}[\rho]}{\delta \rho} \end{aligned}$$

The Kohn-Sham equations are an exact transformation of the original problem (coupled differential equations) in single-particle equations (uncoupled)

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L5 p22 Kohn-Sham 方程: 原问题的精确变换, 误差全在 V_{xc} 。

Hartree-Fock

$$H = -\sum_i \frac{\nabla^2}{m_i} - \sum_{i,j} \frac{(eZ_i) e}{|\vec{R}_i - \vec{r}_j|} + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|\vec{r}_i - \vec{r}_j|}$$

• electrons with kinetic energy

• subject to Coulomb potential of nuclei

• Coulomb-interacting with other electrons

DFT

$$H = -\sum_i \frac{\nabla^2}{m_i} - \sum_{i,j} \frac{(eZ_i) e}{|\vec{R}_i - \vec{r}_j|} + V_H + V_{\text{xc}}$$

• quasi-particles with kinetic energy

• subject to Coulomb potential of nuclei

• interacting with the average, static distribution of all other quasi-particles

• subject to the effect of a mysterious 'exchange-correlation' potential

Much easier maths, but you have to know V_{xc}

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L5 p23 HF 与 DFT 前两项相同, 区别在电子-电子那一项。

Approximation: choose XC functional

One unique expression for $V_{xc}[\rho]$ must exist, but it is unknown.

Local Density Approximation (LDA) **postulates** that:

$$V_{xc}[\rho] = V_{xc}^{LDA}[\rho] = \int \rho(\vec{r}) \varepsilon_{xc}(\rho(\vec{r})) d\vec{r}$$

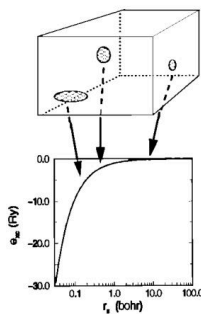
Local Spin Density Approximation (LSDA) **postulates** that:

$$V_{xc}[\rho] = V_{xc}^{LSDA}[\rho_\uparrow, \rho_\downarrow] = \int \rho(\vec{r}) \varepsilon_{xc}(\rho_\uparrow(\vec{r}), \rho_\downarrow(\vec{r})) d\vec{r}$$

Extension: Generalized Gradient Approximation GGA:

$$V_{xc}[\rho] = V_{xc}^{GGA}[\rho, \nabla \rho] = \int \rho(\vec{r}) \varepsilon_{xc}(\rho(\vec{r}), \nabla \rho(\vec{r})) d\vec{r}$$

Meta-GGAs also include second and higher derivatives of the density. This effectively amounts to building more and more accurate Taylor type approximations to the true XC functional.



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L5 p24 LDA → LSDA → GGA → meta-GGA: 逐级加入更多局部信息。

Kinds of DFT functionals

Hybrid Functionals: include some exchange from Hartree-Fock method (e.g. B3LYP)

$$E_{xc}^{B3LYP} = (1-a)E_{xc}^{LSDA} + aE_{xc}^{HF} + b\Delta E_{xc}^{B88} + (1-c)E_{xc}^{LSDA} + cE_{xc}^{LYP}$$

Coefficients a , b , and c are found from fitting some set of experimental data.

Double Hybrid Functionals: include some HF exchange as well as some MP2-like energy (e.g. B2PLYP):

$$E_{xc}^{DHDFT} = (1-a)E_{xc}^{DFT} + aE_{xc}^{HF} + (1-b)E_{xc}^{DFT} + bE_{xc}^{MP2}$$

Coefficients a , b are found from fitting some set of experimental data.

The computation of E_{xc}^{MP2} scales as N^5 , making the calculation quite slow for large systems.

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L5 p30 B3LYP 与双杂化的构成式。

Overview & Performance of DFT

Table 6.1 Perdew classification of exchange–correlation functionals. *Italic* indicates that these include range-separated XC potentials while **bold** indicates inclusion of dispersion.

Level	Name	Variables	Examples	
			Few (≤ 12) parameters	Many (> 12) parameters
1	Local density	ρ	LDA, LSDA, X_{α}	
2	GGA	$\rho, \nabla \rho$	PW91, PBE, BLYP, OLYP, HCTH, KT3	SOGGA11, N12
3	Meta-GGA	$\rho, \nabla \rho, \nabla^2 \rho$ or τ	BR, B95, PKZB, TPSS, τ -HCTH, B97M-V	VSXC, M11L, MN12L
4	Hybrid-GGA	$\rho, \nabla \rho, \nabla^2 \rho$ or τ <i>HF exchange</i>	BHLYP, B3LYP, B3PW91, O3LYP, PBE0, TPSSh, τ -HCTH-hyb, HSE, <i>CAM-B3LYP</i> , <i>ωB97X-V</i>	B97-3, SOGGA11X, <i>M11</i> , <i>N12SX</i> , <i>MN12SX</i> , <i>ωB97X</i> , <i>ωB97XD</i>
5	Double hybrid	$\rho, \nabla \rho, \nabla^2 \rho$ or τ <i>HF exchange</i> <i>Virtual orbitals</i>	B2PLYP, PBE0-DH	

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L5 p36 Jacob 阶梯五级与代表泛函（填表题）。

Magnetic Couplings

$$\hat{H}_{exch} = -J[S_T^2 - S_1^2 - S_2^2]$$

Now, this Hamiltonian can be expanded in the basis of eigenstates of $S=1$ and $S=0$ (as given on previous slide).

$$\begin{aligned} \text{triplet: } E(S_T = 1) &= -J[S_T(S_T + 1) - S_1(S_1 + 1) - S_2(S_2 + 1)] = -J\left[2 - \frac{1}{2} \times \frac{3}{2} - \frac{1}{2} \times \frac{3}{2}\right] = -\frac{1}{2}J \\ \text{singlet: } E(S_T = 0) &= -J[S_T(S_T + 1) - S_1(S_1 + 1) - S_2(S_2 + 1)] = -J\left[0 - \frac{1}{2} \times \frac{3}{2} - \frac{1}{2} \times \frac{3}{2}\right] = +\frac{3}{2}J \end{aligned}$$

Verification:

$$-\frac{1}{2}J \times (2 \times 1 + 1) + \frac{3}{2}J \times (2 \times 0 + 1) = -\frac{3}{2}J + \frac{3}{2}J = 0$$

Now, setting $S = 0$ as ground state (i.e. at 0), we get the energy of the triplet as:

$$E(S_T = 1) = \Delta E_{ST} = -\frac{1}{2}J - \left(+\frac{3}{2}J\right) = -2J$$

If $J > 0 \rightarrow E(S_T = 1) < E(S_T = 0) \rightarrow$ ferromagnetic coupling.

The energy gap between the singlet and triplet is directly linked to the parameter of magnetic interaction.

If $E(S_T = 1)$ and $E(S_T = 0)$ are estimated computational methods, then J could be readily derived. No issues for CI or multireference methods here, except that the correlated calculations are tedious.

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L5 p41 Heisenberg 模型: $\Delta E_{ST} = -2J$, $J > 0$ 铁磁。

Magnetic Couplings from DFT

Can we evaluate the parameter of magnetic interaction using DFT?

Triplet $S=1$:

$$\Psi_{S=1,M=+1} = \alpha(1)\alpha(2)$$

→ one Slater Determinant, suitable for DFT

$$\Psi_{S=1,M=0} = \frac{1}{\sqrt{2}}[\alpha(1)\beta(2) + \beta(1)\alpha(2)]$$

$$\Psi_{S=1,M=-1} = \beta(1)\beta(2)$$

Singlet $S=0$:

$$\Psi_{S=0,M=0} = \frac{1}{\sqrt{2}}[\alpha(1)\beta(2) - \beta(1)\alpha(2)]$$

→ two Slater Determinant, NOT suitable for DFT

As result, the direct DFT might not be suitable for computing the singlet state.

Broken symmetry (BS) approach was proposed to overcome this issue. In BS-DFT, the singlet wave function is not computed, but instead, the energy of one single determinant $\alpha(1)\beta(2)$ is evaluated. Parameter J is derived from the energy difference of two configurations:

$$E_{HS} - E_{BS} = [\langle 1|h|1 \rangle + \langle 2|h|2 \rangle + \langle 12|12 \rangle - \langle 12|21 \rangle]_{HS} - [\langle 1|h|1 \rangle + \langle 2|h|2 \rangle + \langle 12|12 \rangle]_{BS}$$
$$= -\langle 12|21 \rangle = -\left\langle \alpha(1)\beta(2) \left| \frac{1}{r_{12}} \right| \alpha(2)\beta(1) \right\rangle = -2J$$

$$J = \frac{E_{HS} - E_{BS}}{2} = \frac{\langle 12|21 \rangle}{2} = \frac{\left\langle \alpha(1)\beta(2) \left| \frac{1}{r_{12}} \right| \alpha(2)\beta(1) \right\rangle}{2}$$

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L5 p42 BS-DFT: $E_{HS} - E_{BS} = -K_{12}$ 。注意：本页最后一行和 p43 的 J 公式按 p41 的约定符号写反了，正确为 $J = (E_{BS} - E_{HS})/2$ (p42)、 $J = (E_{BS} - E_{HS})/(\langle S^2 \rangle_{HS} - \langle S^2 \rangle_{BS})$ (p43)。详见 [L05 p42-43](#)。

第 6 讲 波函数分析、相对论效应与溶剂化模型

详解：[L06 完整讲义（每页附原图）](#)

必背	一句话
Mulliken 电荷	$Q_A = Z_A - \sum_{\mu \in A} (\mathbf{PS})_{\mu\mu}$: 同原子全给、跨原子平分（讲义 p5 式中的 $\frac{1}{2}$ 多乘了）
Mulliken 的毛病	布居 >2 或负；平分无依据；依赖基组；不重现多极矩
四种电荷	Mulliken/Löwdin（基函数）· Bader（零通量面 $\nabla \rho \cdot \mathbf{n} = 0$ ）· Hirshfeld（自由原子密度比例）
原子电荷 vs 静电势	电荷不是可观测量；MEP 是（红=负=富电子，蓝=正）
键级	$q_{AB} = 2\Sigma\Sigma PS$ （重叠布居）；Mayer 键级更接近 1/2/3
轨道定域化	行列式么正不变；Boys（质心远）· ER（自排斥大）· PM（原子布居集中，保 σ/π ）；定域轨道没有轨道能

相对论	1s 速度 $\approx Z/137\ c$; Au $\gamma \approx 1.22$; s/p 收缩 (直接)、d/f 扩张 (间接)
Dirac 方程	时空都一阶 $\rightarrow 4 \times 4$ 矩阵、4 分量 (大/小分量)、自旋自然出现、正电子态
Pauli 哈密顿量	质量-速度 $-p^4/8m^3c^2$ + SOC $\mathbf{s} \cdot \mathbf{l}$ + Darwin $\delta(\mathbf{r})$; 标量相对论 = 质量-速度 + Darwin
实用相对论	ECP (便宜) · DKH2 / ZORA / X2C (全电子, 需专用基组) ; 1C/2C/4C
溶剂模型	显式 (氢键、贵) · 隐式 PCM (只有 ϵ) · 混合
Born / Onsager	$-(1-1/\epsilon)q^2/2R$; $-[(\epsilon-1)/(2\epsilon+1)]\mu^2/R^3$; $\epsilon > 20$ 后饱和
PCM	原子球空穴 + 表面电荷 $\sigma = -[(\epsilon-1)/4\pi\epsilon]\partial\phi/\partial\mathbf{n}$, 与溶质自治; ΔG = 空穴 + 色散 + 静电
Poisson-Boltzmann	含可移动离子; 线性化 $\rightarrow$ Debye 屏蔽 $e^{-\kappa r}/r$
色散	二阶微扰中两分子同时激发的项 $\rightarrow$ 需双激发 $\rightarrow$ HF/常规 DFT 为零; MP2 需 CP、会高估 π 堆积
UV/Vis	Franck-Condon: 垂直吸收 $>$ 绝热 $>$ 垂直发射; 跃迁偶极为零即禁阻

Mulliken populations and charges

Approximate atomic charges can be derived from the populations of specific atomic orbitals using the procedure suggested by Mulliken. For each atom in the system:

$$Q_A = Z_A - \sum_{\mu \in A} \sum_{\nu \in A} P_{\mu\nu} S_{\mu\nu} - \frac{1}{2} \sum_{\mu \in A} \sum_{\nu \in B} P_{\mu\nu} S_{\mu\nu}$$

This can be expanded as the following set of rules:

- Start with the charge of the bare nucleus.
- If the basis functions ϕ_μ and ϕ_ν are both on atom A, assign all the corresponding density to the charge of atom A.
- If the basis function ϕ_μ is on atom A and ϕ_ν is on atom B, assign half of the corresponding density to the charge of atom A.

Summation without sign and nuclear charge gives *atomic populations*, and summation with spin included gives *atomic spin densities*.

Useful features of Mulliken populations:

- Qualitatively correct (no quantitative method exists anyway).
- Very cheap to calculate (ground state property).
- Show correct trends in chemical reactions.

5

L6 p5 Mulliken 三条规则。注意式中第三项的 1/2 应去掉。

Löwdin populations and charges

Mulliken populations have a few problems:

- Diagonal elements of $P_{\mu\mu}$ may be larger than two, implying more than two electrons in an orbital (Pauli principle violation).
- Off-diagonal elements of $P_{\mu\nu}$ may be negative, implying a negative number of electrons in an orbital.
- Equal partitioning of electron density between participating atoms is without any foundation.
- Diffuse functions centred on atom A are still counted as only belonging to A.
- Mulliken charges do not reproduce the multipole moments of the molecule.

A refinement (at least to points 1-3) was suggested by Löwdin, who used the cyclic permutation property of the trace

$$N = \sum_{\mu,\nu=1}^K P_{\mu\nu} S_{\mu\nu} = \text{Tr}(\mathbf{P}\mathbf{S}) = \text{Tr}(\mathbf{S}^{1/2}\mathbf{P}\mathbf{S}^{1/2}) \quad N_{\mu\nu} = \left[\mathbf{S}^{1/2}\mathbf{P}\mathbf{S}^{1/2} \right]_{\mu\nu}$$

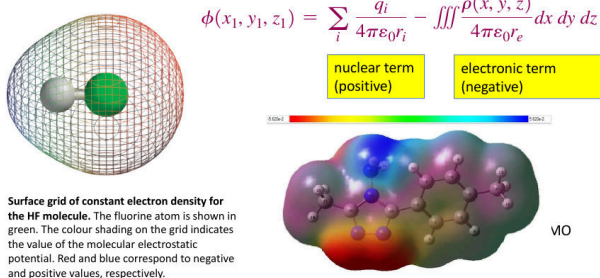
to point out that orthogonalized basis orbitals may be used for the population analysis instead of the (possibly degenerate) original basis set.

6

L6 p6 Mulliken 的五个问题; Löwdin 用 $\mathbf{S}^{1/2}\mathbf{P}\mathbf{S}^{1/2}$ 。

Molecular Electrostatic Potential

The molecular electrostatic potential (MEP) is useful in describing the **charge distribution** of the valence electrons in a molecule.

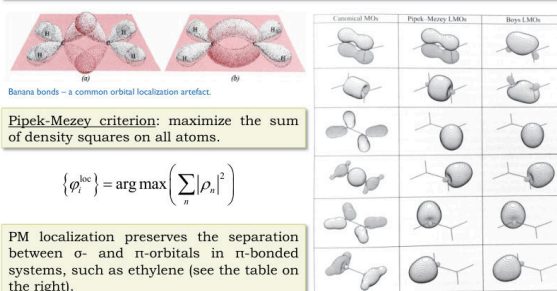


10

L6 p10 静电势 = 核项 (正) - 电子项 (负) ; 红负蓝正。

Molecular orbitals and orbital localization

N.B. Although they still produce the same Slater determinant, the localized orbitals are no longer eigenfunctions of the SCF operator – there is no such thing as the “energy of a localized orbital”.



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L6 p13 定域轨道无轨道能；香蕉键；PM 保留 σ/π 分离。

Relativistic effects

Einstein's relativity theory postulated that:

- Speed of light c is constant in any system of all inertia systems ($\sim 137.036 a. u.$)
- All physical laws are the same in all inertia system $\rightarrow$
- Space coordinates (x, y, z) and time (t) are four coordinates of a single space-time framework, and therefore, are equivalent to each other.
- Lorentz transformations allow to relate the coordinates between two inertia systems

$$\left[-\frac{1}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V \right] \Psi = i \frac{\partial \Psi}{\partial t} \rightarrow \text{Schrodinger equation is NOT relativistically correct: space coordinates are described by second derivative, while time – at first derivative.}$$

$$m = m_0 \left(\sqrt{1 - \frac{v^2}{c^2}} \right)^{-1} \rightarrow \text{the mass of the fast-moving electrons (e.g. 1s) is higher than the mass of valence electrons}$$

This leads to a contraction of the 1s orbital, which in turn, leads to a “lower effective nuclear charge” felt by outer electrons $\rightarrow$ outer orbitals get slightly more diffuse.

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L6 p15 薛定谔方程时空不对称；快电子变重 $\rightarrow$ 收缩。

Energy spectra from Dirac Equation

The four-component Dirac equation can be transformed such as to compute only the Ψ_L (large component, the solution for electrons). The transformed Hamiltonian becomes (Pauli-Dirac equation):

$$\left[\frac{\mathbf{p}^2}{2m} + V - \frac{\mathbf{p}^4}{8m^3c^2} + \frac{\mathbf{Zs} \cdot \mathbf{l}}{2m^2c^2r^3} + \frac{Z\pi\delta(\mathbf{r})}{2m^2c^2} \right] \Psi_L = E\Psi_L$$

Kinetic and potential energy mass-velocity term spin-orbit interaction Darwin correction

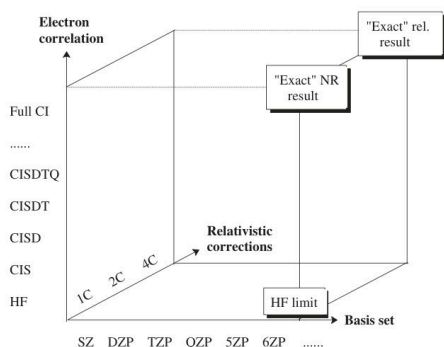
Scalar relativistic corrections = “mass-velocity” + “Darwin” term

Ψ_L is a 2-component wave function (spinor)

Douglas-Kroll Hess is a transformation of the 2-component Dirac Hamiltonian into the non-relativistic + various correction terms

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L6 p18 Pauli 哈密顿量：质量-速度、SOC、Darwin。



Converging the computational results by increasing the basis set, the amount of electron correlation and description of the relativistic effects.

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L6 p22 收敛立方体：基组 $\times$ 相关 $\times$ 相对论。

Continuum solvation effects

We will assume that the molecule sits in a cavity made in a polarisable material. The energy of the molecule has two contributions:

$$E = \langle \Psi | \hat{H} | \Psi \rangle - \langle \Psi | \hat{G} | \Psi \rangle$$

energy of the molecule (incl. electrostatic interactions with the cavity) energy of cavity formation

Table 11.1 Nucleic acid base dipole moments (D) at the SM45.42R/6-31G(d) level

Molecule	Dipole moment		
	On	Chloroform	Water
Adenine	2.6	2.9	3.1
Cytosine	6.5	8.0	8.5
Guanine	5.3	6.7	7.1
Thymine	6.4	7.8	8.2
Uracil	4.5	5.6	6.0

Cavity models:

- Kirkwood-Onsager**: assumes spherical cavity polarized by the dipole moment of the molecule. Completely wrong for all practical purposes.
- Polarisable continuum model (PCM)**: assumes the cavity to be a union of spheres centred on individual atoms. Quite good, but can exhibit numerical issues with tessellation, integration and geometry convergence.
- Isodensity PCM (IPCM)**: uses the SCF isodensity surface as the cavity. Repeats the SCF calculation until the cavity no longer changes.
- Self-consistent isodensity PCM (SCIPCM)**: updates the cavity within the SCF loop, achieving convergence for SCF and cavity at the same time.

F. Jensen, *Introduction to Computational Chemistry*, Wiley, 2007.

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L6 p27 连续介质模型与四种空穴；溶剂让偶极变大。

Obsolete solvent cavity models

Born model refers to a spherical cavity with a charge placed inside:

$$\Delta G_{\text{electr}} = -\frac{\varepsilon - 1}{2\varepsilon} \frac{q^2}{R}$$

Bell model refers to a spherical cavity with a dipole placed inside:

$$\Delta G_{\text{electr}} = -\frac{\varepsilon - 1}{2\varepsilon + 1} \frac{\mu^2}{R^3}$$

Onsager model refers to a spherical cavity with a polarizable dipole:

$$\Delta G_{\text{electr}} = -\frac{\varepsilon - 1}{2\varepsilon + 1} \frac{\mu^2}{R^3} \left[1 - \frac{\varepsilon - 1}{2\varepsilon + 1} \frac{2\alpha}{R^3} \right]$$

Kirkwood model refers to a general multipole expansion of the electrostatic potential of the molecule in a spherical and elliptical (Kirkwood-Westheimer) cavities.

Common feature of all four models: they are completely inaccurate for all practical purposes.

F. Jensen, *Introduction to Computational Chemistry*, Wiley, 2007.

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L6 p28 Born / Bell / Onsager (Onsager 方括号应取倒数)。

Polarizable continuum model

The apparent surface charge is computed as the component of the potential gradient in the internal side of the surface, perpendicular to the surface.

$$\sigma(\vec{r}) = -\frac{\varepsilon - 1}{4\pi\varepsilon} \frac{\partial}{\partial \vec{n}} [\varphi(\vec{r})]_{\text{in}}$$

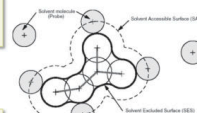


Figure 1. Solvent accessible surface (SAS) traced out by the center of the probe representing a solvent molecule. The solvent excluded surface (SES) is the topological boundary of the union of all possible probes that do not overlap with the molecule.

The resulting charges are fed into the quantum mechanical model (the details are very complicated, see the 2005 review by Tomasi, Mennucci and Cammi for details).

Variations:

1. **Integral equation formalism (IEF) PCM**: (the default method in Gaussian03) uses Green's functions for the Poisson's equation to solve the electrostatic problem.
2. **Isodensity PCM (IPCM)**: uses the SCF isodensity surface as the cavity. Repeats the SCF calculation until the cavity no longer changes.
3. **Self-consistent isodensity PCM (SCIPCM)**: updates the cavity within the SCF loop, achieving convergence for SCF and cavity at the same time.

J. Tomasi, B. Mennucci, R. Cammi, <http://dx.doi.org/10.1021/cr9904009>

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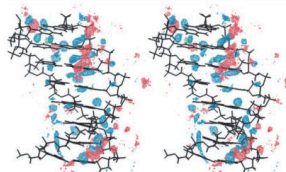
L6 p30 PCM 表观表面电荷; IEF-PCM / IPCM / SCIPCM。

Poisson-Boltzmann model

If we combine the two equations, we get:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \varphi(\vec{r}) = -\frac{ze n_0}{\varepsilon_0 \varepsilon} \left\{ \exp \left[-\frac{ze \varphi(\vec{r})}{kT} \right] - \exp \left[+\frac{ze \varphi(\vec{r})}{kT} \right] \right\}$$

This is known as the Poisson-Boltzmann equation. Because it assumes the presence of independent point charges in the liquid, it is *not applicable to dielectric liquids*.



Cross-eyed stereo plot of water (blue), sodium (red) and chloride (green) ion distribution around a DNA molecule.

$$\Delta G_{\text{electr}} = \frac{1}{2} \int \rho(\vec{r}) \varphi_{\text{vac}}(\vec{r}) d^3\vec{r}$$

$$\varphi_{\text{vac}}(\vec{r}) = \varphi_{\text{solv}}(\vec{r}) - \varphi_{\text{vac}}(\vec{r})$$

The equation is solved numerically on a 3D grid. The resulting reaction potential (the difference between the solvated and the vacuum potential) determines the electrostatic part of the Gibbs free energy of solvation.

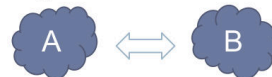
D.J. Shaw, *Colloid and Surface Chemistry*, Butterworth Heinemann, 2003.

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L6 p35 Poisson-Boltzmann 方程; DNA 周围的离子氛。

Non-covalent interactions

Van der Waals interaction and aromatic stacking are due to the two molecules polarizing each other on approach:



$$\hat{H} = \hat{H}_A + \hat{H}_B + \hat{H}_{AB}; \quad \hat{H}_{AB} = \sum_{a \in A} \sum_{b \in B} \frac{z_a z_b}{r_{ab}}$$

$$E = E_A + E_B + \underbrace{\langle 0_A 0_B | \hat{H}_{AB} | 0_A 0_B \rangle}_{\text{simple electrostatics - not good enough}} + \underbrace{\sum_{nk} \frac{\langle 0_A 0_B | \hat{H}_{AB} | n_A k_B \rangle \langle n_A k_B | \hat{H}_{AB} | 0_A 0_B \rangle}{E_0^{(A)} + E_0^{(B)} - E_k^{(A)} - E_n^{(B)}}}_{\text{requires the knowledge of all excited determinants}}$$

Unless the excited determinants are introduced accurately in a particular method, it is unlikely to yield accurate Van der Waals interactions.

F. Jensen, *Introduction to Computational Chemistry*, Wiley, 2007.

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L6 p37 分子间作用: 一阶静电 + 二阶 (诱导、色散需双激发)。

UV/Vis spectra

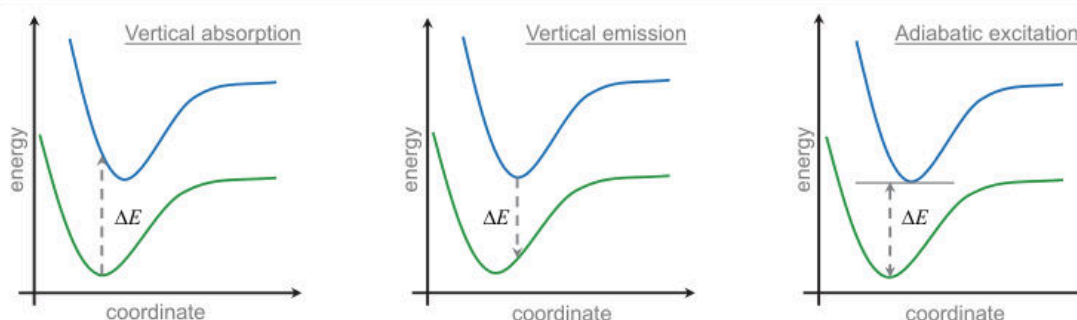
Accurate vertical excitation energies are usually obtained from configuration interaction calculations (excited determinants are expensive):

$$\Delta E_k = \langle \Psi_k | \hat{H} | \Psi_k \rangle - \langle \Psi_0 | \hat{H} | \Psi_0 \rangle$$

and the transition probabilities are calculated using perturbation theory:

$$\vec{d}_{kn} = \langle \Psi_k | \hat{\vec{d}} | \Psi_n \rangle \quad \hat{\vec{d}} = e\vec{r}$$

The energy minima for the ground and the excited states do not necessarily match:



F. Jensen, *Introduction to Computational Chemistry*, Wiley, 2007.

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L6 p39 垂直吸收 / 垂直发射 / 绝热激发。

方法对比总表 (跨第 2-5 讲)

方法	标度	变分?	尺寸一致?	静态相关	动态相关	什么时候用
HF	N^4	✓	✓	✗	✗	参考态; 单独用意义不大
DFT (杂化)	N^{3-4}	近似泛函不严格	✓	✗	近似 ✓	机理、大体系主力
MP2	N^5	✗	✓	✗ (会发散)	✓	便宜的从头算相关
CISD	N^6	✓	✗	✗	✓	基本被 CCSD 淘汰
CCSD(T)	N^7	✗	✓	✗ (会崩)	✓✓	黄金标准、做基准

CASSCF	活性空间 阶乘	✓	✓ (空间选 对)	✓	X	多参考体系 (后接 PT2)
CASPT2	—	X	近似	✓	✓	过渡金属、镧系、 激发态
FCI	$N!$	✓	✓	✓	✓	3-4 原子基准

记忆口诀：变分的都是"直接最小化能量"的 (HF、CI、CASSCF)；CC 和 MP 靠投影/微扰，所以不变分；只有截断 CI 不尺寸一致。

符号撞车对照 (最容易看混的地方)

符号	在哪里是什么意思
φ / χ	L2/L3: φ = 分子轨道, χ = 基函数; L5 p4 反过来: χ = 轨道, φ = 基函数
D / P	都是密度矩阵: L3 叫 D , L5 和 Gaussian 输出叫 P
J	① J_{ij} 库仑积分 (L2-L3) ② J 磁交换耦合常数 (L5 p40) ——完全不同
K	交换积分 K_{ij} ; BS-DFT 中 $E_{\text{HS}} - E_{\text{BS}} = -K_{12}$
α	① α 自旋 ② 高斯指数 $e^{-\alpha r^2}$ ③ 变分参数 (L3 p29) ④ 极化率 (L5 p44) ⑤ 基函数下标 (L3 p54)
ϵ	① 轨道能 ② ϵ_0 真空介电常数 ③ ϵ_{xc} 每电子 XC 能量密度 (L5 p24) ④ 溶剂相对介电常数 (L6 PCM) ⑤ 水模型 LJ 势阱深度 ϵ_0 (L6 p24)
S	① 重叠矩阵 $S_{\mu\nu}$ ② 总自旋 S ③ 单激发下标 (L4 p12、L5 p6)
T	① 动能 ② 温度 ③ CC 簇算符 ④ 三激发下标 ⑤ 三重态
(n)	上标括号 = 微扰阶数 (L4 p29), 不是幂次
$[]$ vs $()$	$E[\rho]$ 方括号 = 泛函 (输入是函数); $f(x)$ 圆括号 = 函数 (输入是数)

高频概念题 (一句话答案)

问题	答
HF 丢了什么?	电子相关: 平均场里电子看不到彼此的瞬时位置
J 和 K 的区别	J 经典库仑、所有对都有; K 纯量子、仅同自旋、降能量

为什么三重态低于单重态	同自旋空间波函数反对称 → 电子不重合 → 排斥小 ($-2K$)
Roothaan 为什么有 S	原子轨道基函数不正交
HF 为什么要迭代	Fock 算符依赖轨道本身
RHF 为什么 H_2 解离错	单行列式锁死 50% 离子项
静态 vs 动态相关	静态: 参考态本身错 (简并) → CASSCF; 动态: 细节缺失 → MP2/CC
为什么 CISD 不尺寸一致	缺两片段各双激发的四激发交叉项
为什么 CC 尺寸一致	$e^{A+B} = e^A e^B$, 波函数自动分解
MP2 为什么无单激发	Brillouin 定理
MP2 什么时候崩	近简并, 分母趋零
为什么用 GTO	高斯乘积定理 → 积分解析
BSSE 是什么	基函数互借 → 结合能偏大; Counterpoise
Koopmans 的假设	冻结轨道; HF 靠误差抵消, DFT 不适用
HK 两个定理	① ρ 唯一决定 V_{ext} ② $E[\rho]$ 在 ρ_0 处最小
KS 的核心思想	无相互作用替身体系重现同一个密度, 未知全放进 V_{xc}
DFT 是 ab initio 吗	原理上是; 实用泛函含经验参数, 严格说不是
自相互作用误差	近似 E_{xc} 不能抵消 J_{ii} → 过度离域
DFT 为什么没色散	泛函局域, 看不到远处关联涨落 → D3
为什么要 BS-DFT	开壳层单重态要 2 个行列式, KS 只有 1 个
变分原理的限制	只保证能量上界, 不保证其他性质与能量差
虚频的意义	0 = 极小点, 1 = 过渡态